DATA SCIENCE LABORATORY MANUAL

Module II ? Python for Data Science

Topic: Python Basics

OVERVIEW & LEARNING OBJECTIVES

Core syntax, data structures (lists, tuples, dicts), functions, list comprehensions, and error handling.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Python Basics in modern Data Science workflows.
- Detailed breakdown: Core syntax, data structures (lists, tuples, dicts), functions, list comprehensions, and error handling.
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.